

# Soft Spaces Phase 4

*Interim Summary: Application Advantage, Circuit Robustness, and Breakpoint Localization*

Scientific checkpoint - 15 September 2026

**Executive conclusion.** Phase 4 has moved Soft Spaces from a structural Phase-3 observation to a concrete operational result. Within the frozen 8-qubit model, REAL Soft-Space information provides a reproducible ranked-search advantage over matched NULL controls. The advantage survives controlled score noise, a hardware-oriented noise proxy, and an explicit noisy Qiskit/Aer circuit channel through CX=400. Exploratory high-CX tests indicate that the Precision@8/search-efficiency advantage is weakened around CX=800 and approaches breakdown around CX=1200.

## 1. Objective

Phase 4 asks whether the structured regions discovered in Phase 3 can be used operationally. The first application is cross-energy ranked search: response information on the -E side is used to prioritize targets on the +E side. REAL is compared with a matched NULL reference, and target-side information is not used to construct the ranking.

## 2. Frozen framework

| Item              | Definition                    |
|-------------------|-------------------------------|
| System            | 8 qubits                      |
| Hamiltonian       | 11-term random-Pauli model    |
| Phase-3 kernel    | v30.7 / namespace v30.2       |
| Independent seeds | 25043000-25043011             |
| Comparison        | REAL vs matched NULL          |
| Primary metric    | Precision@8                   |
| Secondary metrics | Target evaluations and NDCG@8 |

## 3. Evidence chain

### v40.0 - Application screening

The first application test showed that Phase-3 structure can be converted into a ranked-search advantage.

| Metric            | Result                 |
|-------------------|------------------------|
| REAL Precision@8  | 1.000000               |
| NULL Precision@8  | 0.822917               |
| Delta Precision@8 | +0.177083              |
| 95% bootstrap CI  | [+0.125000, +0.229167] |
| Sign test         | p = 0.00048828         |
| Wilcoxon          | p = 0.00097609         |
| REAL evaluations  | 8.000000               |
| NULL evaluations  | 10.916667              |
| Decision          | PASS                   |

### v40.1 - Controlled source-score noise

The REAL advantage remained statistically supported under substantial controlled corruption of the source ranking. The primary checkpoint  $\eta=1e-3$  passed all criteria, and the predefined robustness gates continued to pass through  $\eta=0.4$ .

### v40.2 - Hardware-oriented proxy

Finite shots, readout error and depolarizing-noise/depth effects were introduced. The application advantage remained positive through the largest tested effective two-qubit depth,  $N_{2q}=400$ . This remained a proxy rather than an explicit quantum-circuit simulation.

### v40.3b - Explicit Qiskit/Aer circuit test

v40.3b moved the signal through an explicit noisy Qiskit/Aer circuit channel using finite shots, readout error, one- and two-qubit depolarizing noise, and deliberate CX identity pairs.

| CX  | REAL P@8 | NULL P@8 | Delta P   | REAL eval | NULL eval | Decision |
|-----|----------|----------|-----------|-----------|-----------|----------|
| 0   | 0.978021 | 0.825729 | +0.152292 | 8.240     | 10.938    | PASS     |
| 10  | 0.979427 | 0.825833 | +0.153594 | 8.227     | 10.956    | PASS     |
| 20  | 0.978333 | 0.825052 | +0.153281 | 8.240     | 10.965    | PASS     |
| 50  | 0.977031 | 0.825625 | +0.151406 | 8.263     | 10.953    | PASS     |
| 100 | 0.975990 | 0.826302 | +0.149688 | 8.267     | 10.962    | PASS     |
| 200 | 0.974844 | 0.825990 | +0.148854 | 8.290     | 10.973    | PASS     |
| 400 | 0.967292 | 0.825990 | +0.141302 | 8.396     | 10.977    | PASS     |

**Primary checkpoint at CX=20:** Delta Precision@8 = +0.153281, 95% CI [+0.102448, +0.208281], sign-test  $p = 0.00048828$ , Wilcoxon  $p = 0.00097609$ . Decision: PASS.

## 4. Breakpoint localization

Because no breakdown occurred through CX=400, a fast high-CX locator was used to identify the transition region before spending more time on explicit Aer simulations.

| CX   | REAL P@8 | NULL P@8 | Delta P   | Classification |
|------|----------|----------|-----------|----------------|
| 800  | 0.899687 | 0.854531 | +0.045156 | WEAK           |
| 1200 | 0.858125 | 0.846562 | +0.011563 | NEAR-BREAKDOWN |
| 2000 | 0.717656 | 0.728594 | -0.010938 | NEAR-BREAKDOWN |
| 3000 | 0.562813 | 0.568281 | -0.005469 | NEAR-BREAKDOWN |
| 4000 | 0.516094 | 0.520156 | -0.004062 | NEAR-BREAKDOWN |

## 5. Explicit Aer spot-check around the transition

A deliberately lightweight explicit Aer spot-check tested CX=800 and CX=1200 using two seeds, 10 repetitions and 512 shots.

| CX   | REAL P@8 | NULL P@8 | Delta P   | REAL eval | NULL eval | Delta NDCG |
|------|----------|----------|-----------|-----------|-----------|------------|
| 800  | 0.818750 | 0.731250 | +0.087500 | 10.150    | 12.350    | +0.192527  |
| 1200 | 0.768750 | 0.762500 | +0.006250 | 13.950    | 12.250    | +0.230355  |

The explicit circuit simulation qualitatively confirms the fast locator: the Precision/search-efficiency advantage is still visible at CX=800 but is nearly gone at CX=1200. NDCG remains favorable to REAL at CX=1200, suggesting that some ranking structure survives beyond the point where Top-8 Precision ceases to be operationally useful.

## 6. Current interpretation

The evidence supports a gradual degradation rather than an abrupt binary failure. Within the present circuit encoding and noise model, the application advantage is strong through CX=400, weakened around CX=800, and close to operational breakdown around CX=1200.

This supports describing Soft Spaces, at the current application level, as a response-guided, problem-adaptive Hilbert-space prior: the physical response is used to identify regions that should be searched first rather than searching uniformly.

## 7. Established and not yet established

| Status              | Claim                                                                              |
|---------------------|------------------------------------------------------------------------------------|
| Established         | REAL provides a reproducible ranked-search advantage in the frozen 8Q model.       |
| Established         | The advantage survives source-score noise and hardware-oriented noise models.      |
| Established         | The advantage survives an explicit noisy Qiskit/Aer circuit through CX=400.        |
| Established         | Exploratory tests localize operational breakdown approximately around CX=800-1200. |
| Not yet established | General validity for arbitrary Hamiltonians or arbitrary quantum programs.         |
| Not yet established | Native execution of the full Soft-Spaces Hamiltonian/projector response on a QPU.  |
| Not yet established | Demonstration on real IBM hardware.                                                |
| Not yet established | Quantum advantage in the computational-complexity sense.                           |

## 8. Recommended next step

No additional high-resolution breakpoint study is required at the present stage. The current Phase-4 application checkpoint is sufficiently mature to freeze and commit. The next scientifically distinct step should be preparation for a real-hardware experiment on a suitable connected 8-qubit subgraph of an IBM Heron processor, followed, if feasible, by replication on more than one device.

## 9. Interim conclusion

Phase 4 has reached a meaningful scientific checkpoint. Soft-Space structure has been converted into a measurable application-level ranked-search advantage, and that advantage remains robust through an explicit noisy circuit simulation. The present evidence is sufficient to archive the current result, make a clean repository commit, and move the work toward physical-hardware validation.